

Grammaticalization of sentence adverbs and modal particles revisited

Katrin Axel-Tober¹, Marco Coniglio², Kalle Müller¹, Katharina Paul²

¹University of Tübingen, ²University of Göttingen

Abstract

This paper investigates the development of sentence adverbs (SAdvs) and modal particles (MPs) by especially focusing on the German language. By exploring their common grammaticalization paths, we claim that these are much more restricted than previously assumed. In particular, we argue that two pathways seem to play a major – if not exclusive – role, namely one starting from lower adverbs and one from parenthetical clauses, and that syntactic ambiguity in the middle field plays a fundamental role in this process. In addition, we show that there is no evidence of secondary grammaticalization occurring within the class of sentence adverbs and modal particles.

1. Introduction

In this paper, we investigate the development of sentence adverbs and modal particles. We will highlight their common paths of development and examine the role of syntactic ambiguity in this process. Our focus is on German which exhibits a rich inventory of such items.

From a syntactic point of view, SAdvs and MPs have been described as elements in the higher functional domain of the clause (cf. Cinque 1999). As to their function, SAdvs express the speaker's attitude with respect to the evaluation, certainty, or information source (1) of/about the core proposition, while MPs (e.g. *denn, ja, vielleicht, eigentlich, aber*) express use-conditional or expressive meaning (cf. Gutzmann 2015) and sentence type dependency (2).¹

- (1) Elif kauft sich **vielleicht** / **glücklicherweise** / **anscheinend** ein Haus.
 Elif buys herself maybe fortunately apparently a house
 'Elif will perhaps/fortunately/apparently buy a house.'

¹ In descriptive and theoretical accounts on German, it is generally assumed that the category of MPs is characterized by some clearly distinguishable syntactic and prosodic properties that set them apart from adverbs. For example, MPs are restricted to the middle field, cannot be modified or coordinated, are usually unaccented and are dependent on sentence type. It should be noted, however, that except for the middle-field restriction, these characteristics also apply to most items from the special subclass of SAdvs – in contrast to 'ordinary' VP-/predicate-related adverbs. Cf. also Abendroth Scherf (2018) for similar observations on Swedish MPs. Outside Germanic, Romance MPs have received some attention in the literature (e.g. Remberger 2020), but here the demarcation from SAdvs is even less clear.

(2) A: Elif hat sich ein Haus gekauft.

Elif has herself a house bought

‘Elif has bought a house.’

B: Hat die *vielleicht/aber* gute Nerven! Bei den hohen Zinsen!

Has she PRT good nerves with the high interests

‘Wow, that is really ($\neq$ perhaps / $\neq$ however) courageous of her given the high construction interests!’

Was hat sie *eigentlich/denn* für einen Job?

What has she PRT for a job

‘What kind of job does she have?’

$\approx$ ‘actually’

$\approx$ ‘I am somewhat surprised that she can afford this and this is why I ask what job she has.’

With a focus on semanto-pragmatic factors (cf. Traugott 2010, Diewald 2011), the diachronic developments of both SAdvS and MPs have been described in different terms in the literature. Initially, such changes were considered as cases of grammaticalization, but as one can immediately notice, this poses a challenge for the grammaticalization parameters discussed in the seminal work of Lehmann (1995), since apart from the loss of syntactic freedom (only affecting MPs, which are confined to the middle field), the applicability of all other parameters (obligatorification, decrease of scope/condensation, paradigmaticization, loss of phonological and semantic integrity) is not as straightforward when applied to MPs and SAdvS.

In contrast, other authors refer to the notion of (inter-)subjectification as a distinct and autonomous semanto-pragmatic process, in which “[m]eanings tend to become increasingly based in the speaker’s subjective belief state/attitude toward the proposition.” (Traugott 1989: 34–35; 2010). For example, this notion was applied to English adverb(ial)s such as *obviously*, *indeed*, *in fact* etc. (ibid.) or to German MPs (e.g. Autenrieth 2002).

More recently, the idea of grammaticalization has been revived under the more specific notion of pragmaticalization, which is defined as grammaticalization of discourse functions by Diewald (2011) (cf. Detges & Waltereit 2016). For our purpose, possible discourse functions for SAdvS and MPs might be e.g. speech act qualifications, hedging, emotional attitudes and the need to relate the utterance to the common ground.

Note that different categories have been postulated as sources in the developments of these elements, especially MPs are often assumed to go back to a variety of different source categories such as adjective-adverbs, low adverbs, SAdvS, focus particles and conjunctions (cf. Burkhardt 1994, Autenrieth 2002, Diewald 2011). However, this picture might be deceptive due to the large synchronic heterosemy of these elements.

In this paper, we argue (mostly based on German) that the source categories and paths of development are much more restricted than previously assumed (cf. Axel-Tober & Müller 2017, Coniglio 2022). In section 2, we identify two major paths: SAdvS and MPs have primarily developed from lower adverbs (2.1) or from parenthetical clauses (2.2). In sections 3 and 4, we show that these paths are determined by clear syntactic constraints, which have been mostly neglected up to now, due to the major focus on pragmatic aspects (but see van Gelderen 2011 for an important exception). In doing so, we intend to show the limits of existing theories and to focus on the observable syntactic sources and paths and on the important generalizations that we draw from them. We will thus also address the two leading questions: what is the relation between form and interpretation in pragmaticalization phenomena? And, ultimately, which role does syntactic reanalysis play?

2. Two major paths

2.1 Path I: Lower event-related adverbials > SAdvS/MPs

Manner or temporal adverbs or adverbial phrases have often been noted to serve as sources in grammaticalization processes leading to SAdvS and MPs (e.g. Wegener 2002, Autenrieth 2002 for German MPs). A hallmark example is the development of the German suffix *-erweise*, which goes back to the combination of the genitive inflection of an adjective and the noun *Weise* (OHG *wīsa* ‘manner, form of appearance’), a cognate of English *wise* (Paraschkewoff 1976, Pittner 2015).

- (3) Also ward si mit gewalt aus den armen ihrer mutter här-aus-gerissen /
 also was she with force out the arms of her mother away-jerked
 welche ihr ganz *erbärmlicher weise* nach-ruhf / ach! meine tochter /
 who her very pitifully after-shouted oh my daughter

‘She was violently jerked away out of her mother’s arms, who shouted very pitifully after her: Oh! My daughter!’

(DTA, Ritterhold von Blauen [i. e. Zesen, Philipp von]: Adriatische Rosemund. Amsterdam, 1645. #301)

In (3), *erbärmlicher weise* is a genitive DP consisting of the adjective *erbermlich* ‘miserable’ and the noun *Weise* ‘manner’. It functions as a manner adverbial: ‘in a pitiable way, pitifully, miserably’. Later, *-erweise* developed into a suffix(oid). The formation [ADJ+*erweise*]_{ADV} has been a productive pattern since the 16th century. In present-day German, however, the derivational suffix *-erweise* with the manner interpretation is only productive with the present participle stem of a verb (cf. Elsner 2015). With ‘ordinary’ adjectival stems, *-erweise* adverbs convey an evaluative interpretation (4) and thus they exemplify the path ‘lower event-related adverb > SAdv’.²

- (4) ***Erbärmlicherweise*** glaubt der Stadtrat, dass Lebensqualität allein
it.is.a.pity believes the city-council that life’s-quality only
an Fress- und Saufbuden [...] festzumachen ist.
at food- and drink-booths to.determine is
‘The city councillor believes that life quality comes from food and drink booths, which is a pity’
(DeReKo, Nürnberger Nachrichten, 01.10.2007)

In the following, we will demonstrate that the path leading from a lower adverb (in particular from a manner or temporal adverb) to a SAdv or MP is even more prominent than previously assumed. It also carries over to cases that have been traced back to other source categories (notably to high adverbs and conjunctions) in parts of the literature.

Another example is the grammaticalization of German *eigentlich* from the MHG adjective *eigenlich* ‘proper, peculiar’ (with a later epenthetic *-t-*), which – besides being still used as an adjective – also occurs both as a SAdv and a MP in present-day German (cf. Coniglio 2022: 22). As an adjective in adverbial use, *eigentlich* originally was typically located in a relatively low syntactic position, e.g., after the negation:

² In their cross-linguistic study on SAdvS in the languages of Europe, Ramat & Ricca (1998: 204) identify *-erweise* and its cognates in Dutch, Swedish, Danish and Swiss German as the only derivational suffix that is used to derive SAdvS and nothing else.

- (5) ob etwas weiters von Leichnamen nutzlichs geblieben /
 if anything further of corpses useful remained
 darff ich nit *eigentlich* sagen
 can I not in.a.proper.way say
 ‘if anything useful of corpses further remained, I cannot say in a proper way’
 (DTA, Andreaä, Johann Valentin: Chymische Hochzeit Christiani Rosencreutz Anno
 1459. Straßburg, 1616. #118)

Nonetheless, low adverbs could (and can still) be moved to a position preceding the negation in German. In this position, the low adverb *eigentlich* is syntactically ambiguous with the MP *eigentlich*, which obligatory preceded the negation:

- (6) Mond ist *eygentlich* nicht vrsach der bewegung des Meeres.
 moon is actually/properly not cause of.the movement of.the sea
 ‘The moon is actually/properly not the cause of the movement of the sea.’
 (DTA, Arndt, Johann: Vom wahren Christenthumb. Bd. 4. Magdeburg, 1610. #172)

A third example of this path is provided by the development of German *vielleicht*, which, like *eigentlich* above, can occur as an epistemic SAdv (7a) or a MP (7b) in present-day German:

- (7) a. Er ist *vielleicht* in Rom.
 ‘Maybe, he is in Rome.’
 b. Er ist *vielleicht* frech!
 ‘He is really impertinent!’
 (Coniglio 2022: 23)

It is commonly assumed that the MP *vielleicht* has developed very late in NHG from the homophonous SAdv. However, this path does not explain the assertive character of the MP. Therefore, a better account locates the source of this grammaticalization process not in the SAdv *vielleicht*, but rather in the MHG adverbial phrase *vil lîhte(r)* ‘very easily, with certainty’, shown in (9) (Coniglio 2022: 23):

- (9) [...] daz tuot mir nû *vil lîhte* wê

this does to.me now very easily woe
 ‘this hurts me now very easily’
 (MWB, Reinmar der Alte, MF:Reinm 24: 4, 6, 12th c.)

Given an appropriate context, the adverbial phrase *vil lihte(r)* easily allows a reanalysis to a MP meaning ‘really’ (cf. Coniglio 2022: 24):

- (10) [...] sinnelôser Thêodâ, / dû bist ***vil lihte*** âne vernunst, /
 foolish Theoda you are PRT without reason
 ‘Foolish Theoda, you are really deprived of reason!’
 (MWB, Rudolf von Ems, RvEBarl 12995, ca. 1220)

If these observations are on the right track, the SAdv *vielleicht* and the homophonous MP must have developed independently from the same adverbial phrase. The MP is thus probably much older than assumed so far (Coniglio 2022: 23–24).

2.2 Path II: Clausal parentheticals > SAdv/MPs

A second major grammaticalization path leading to the emergence of SAdv (and possibly of MPs) is one starting from parenthetical sentences. These structures are first realised as independent sentences, which are used “parasitically” to a main clause. At a further stage, they are reanalysed as adverbs belonging to the matrix sentence in which they occur. Such developments seem to have affected or to be affecting epistemic or evidential elements such as English *maybe*, French *peut-être*, German *glaub* (lit. ‘(I) think/believe’), Afrikans *glo* (cf. Thompson & Mulac 1991) and Ital. *credo* (Giorgi & Pianesi 2005).

As argued in Axel-Tober (2016), some speakers, in particular of southern German, use the expression *scheints* as an evidential SAdv. *Scheints* is a reduced and unverbated form of a parenthetical verb-first (= V1) clause: *scheint es* (lit. ‘seems it’). That these speakers no longer use *scheints* only as a parenthetical, but also as a fully-fledged (sentence) adverb is evidenced by the fact that they can topicalize it to the SpecC position even in the written language (e.g. in cross-regional newspapers) (11). The prefield is not accessible to parenthetical expressions.³

³ In German, the SpecC position has to be filled via movement of an XP from the middle field (or insertion of an expletive) in canonical declaratives. Parentheticals cannot satisfy this constraint:

- (11) *Scheints* hat das so manche [...] nicht kapiert.
 apparently has that so some not understood
 ‘Apparently, some have not understood that.’
 (DeReKo, spektrumdirekt, 18.01.2006)

Note that English SAdvs have been analysed as parenthetical constructions both in syntactic (e.g. Jackendoff 2002, Alexiadou 1997) and semantic (e.g. Potts 2005) accounts. While semantically, this type of analysis is applicable to evaluative, but not to epistemic and evidential SAdvs (Müller 2021),⁴ even evaluative SAdvs are undoubtedly syntactically integrated and behave like ‘ordinary’ constituents.⁵ As such, they can occur in the prefield just like any other type of adverb (temporal, causal etc.):

- (12) a. *Vielleicht / Erbärmlicherweise / %Scheints* hat sie das nicht kapiert.
 perhaps which.is.a.pity apparently has she that not understood
 ‘Perhaps, she has not got the message./ She has not got the message, which is a pity.
 / it seems’
 b. *Gestern / Daher* hat sie es nicht kapiert.
 Yesterday / therefore has she it not understood.
 ‘Yesterday/ Therefore, she didn’t get it.’

3. Syntactic ambiguity and reanalysis in the middle field

-
- (i) *Sagt die Lehrerin/ Glaube ich/ Scheint es hat sie das nicht kapiert.
 says the teacher / believe I / seems it has she it not understood
 ‘The teacher says / I believe / It seems that she didn’t get that.’

⁴ More specifically, Pott’s (2005) framework distinguishes between two different levels of semantic content, of which the first level is completely independent of the second level. Since the use of epistemic and evidential expressions results in a lower commitment of the speaker to the main proposition, purely treating them on this separate second level would yield the wrong predictions for sentences like *Maybe [it’s raining], maybe [it’s not]*: the conjunction between the main parts (in brackets) of this sentence would yield a contradiction. This contradiction can be lifted by epistemic SAdvs like *maybe*, but not by evaluative ones.

⁵ There is a very small number of grammaticalized speech-act adverbs in German. Items from this class cannot occur in SpecC, they have to be placed further to the left and are prosodically set off from the host clause. This is why they are not classified as adverbs in most descriptive grammars of German.

- (i) Kurzum: sie hat es nicht kapiert.
 In.short she has it not understood
 ‘In short, she didn’t get it.’

In the last section, we discussed two major paths as central for the development of SAdvS and MPs:

- (i) Lower event-related adverbs > MPs/SAdvS
- (ii) Clausal parentheticals > SAdvS (/MPs?)

In order to account for these facts and in contrast to semanto-pragmatically-driven accounts, we intend to emphasize the syntactic ambiguity and reanalysis – and therefore the often neglected role of syntax in general – as crucial factors in the grammaticalization paths of both types of elements (cf. Roberts & Roussou 2003, van Gelderen 2011).⁶ Specifically, we argue that the two paths share a common syntactic condition: syntactic contexts with ambiguous surface orders in combination with interpretative ambiguity enabled the reanalyses of both low adverbs and parentheticals into SAdvS or MPs.

In the German middle field, manner adverb(ial)s and sentence adverb(ial)s may show up in the same surface position:

- (13) a. Das Clownspaar Wenzel & Mensching hatte [...] den Abschied
the clowns.duo Wenzel & Mensching had the farewell
von diesem nunmehr als total absurd empfundenen Leben
from this now as totally absurd felt life
lachenderweise erleichtert.
laughing-wise up.lightened
‘The clowns duo Wenzel & Mensching had lightened up the farewell to this life –
which was now felt as totally absurd – with laughter.’
(DeReKo, Berliner Morgenpost, 08.10.1999, S. 35)
- b. Das Clownspaar Wenzel & Mensching hatte [...] den Abschied
the clowns.duo Wenzel & Mensching had the farewell
von diesem nunmehr als total absurd empfundenen Leben
from this now as totally absurd felt life
glücklicherweise erleichtert.

⁶ While reanalysis is commonly assumed as a mechanism of language change (cf. e.g. Hopper & Traugott 1993, Harris & Campbell 1995, Lehmann 1995, Traugott 2011), the status of reanalysis within grammaticalization or even language change in general has been disputed (cf. e.g. Haspelmath 1998, Fischer 2011).

fortunately up.lightened

‘Fortunately, the clowns duo Wenzel & Mensching had lightened up the farewell to this life – which was now felt as totally absurd.’

(based on a)

In (13), the manner adverb *lachenderweise* (present participle of *tanzen* + *-erweise*)⁷ and the SAdv *glücklicherweise* ‘luckily, fortunately’ (> adjective *glücklich* + *-erweise*, now lexicalized) show up relatively far right in the middle field following the object. In an early study on German SAdvs, Hetland (1992) proposes that they can be freely base-generated as adjuncts in their surface positions – due to quantifier raising at LF, their (potentially low) syntactic positioning thus do not affect interpretation. More recent analyses take SAdvs to be merged in fixed high positions, either as adjuncts to IP (Frey & Pittner 1998) or in dedicated functional projections in a high functional domain in the left region of the IP/TP or above it (cf. Coniglio 2011: 106 for a Cinquean analysis of German; Krifka to appear for a recent different cartographic account). In such accounts, occurrences of high adverbs in an apparently far right position result from object scrambling, as possibly in (13b).

As mentioned above, diachronically the manner interpretation of *(-er)weise*-adverbs preceded the evaluative/sentence-adverbial use. The evaluative adverb plausibly developed in ambiguous bridging contexts where both the original manner and the new evaluative interpretation were possible:

(14) sonstn weren wir *erbermlicher weise* zertreten worden

a. ‘... else we would have been crushed in a miserable way’

b. ‘... else we would have been crushed, which would have been miserable’

(DTA, Rollenhagen, Gabriel: Vier Bücher Wunderbarlicher biß daher vnerhörter/ vnd vngleublicher Indianischer reysen. Magdeburg, 1603. #52)

In (14), *erbermlicher weise* has two possible readings: on the one hand, it can be understood as a genitive DP consisting of the adjective *erbermlich* ‘miserable’ and the noun *Weise* ‘manner’ (14a). This manner adverbial – assumedly the original construction – would occur in a low IP-internal position. On the other hand, *erbermlicher weise* can be interpreted as an evaluative adverb composed of the adjective *erbermlich* ‘miserable’ and the suffix(oid) *-erweise* (14b). As

⁷ As mentioned above, only *-erweise* adverbs based on present participles are still productive as manner adverbs in German (Elsner 2015).

a SAdv, it occupies a high position and the weak subject pronoun *wir* must have scrambled or moved to the Wackernagel position further to the left. The (latter) formation [ADJ+*erweise*]_{ADV} has been a productive pattern since the 16th century. We thus hypothesize that the upward reanalysis and recategorization (manner adverb > SAdv) must have occurred before that period.

Reanalysis in the middle field due to syntactic and interpretative ambiguity has also been argued to have fuelled the development of further types of SAdvS in German (cf. Axel-Tober 2016, Axel-Tober & Müller 2017 on evidentials and epistemics). It could be objected that the postulated type of reanalysis based on ambiguity in the middle field is too language-specific confronted with the fact that SAdvS go back to lower adverbs also in many other languages (e.g. Swan 1988 on English and Ramat & Ricca 1998). Note, however, that further Germanic and Romance languages had more flexible word orders in earlier phases. German provides an accessible window to the diachronic mechanisms of adverbial reanalysis as it has retained its flexibility in the middle field until today.

In the case of adverbs and – possibly – MPs of parenthetical origin, we postulate a parallel type of reanalysis. However, in this case the syntactic ambiguity does not arise from flexible middle-field order due to scrambling, but from the freedom in parenthetical niching. It is a well-known fact that German parenthetical clauses can occur in any surface position in the middle field (Reis 2002) and that they may show up in the same surface positions as MPs or SAdvS. V1 parentheticals of the type *sagt/denkt/antwortet sie* ‘says/thinks/answers she’, *scheint es* ‘seems it’ or *glaub (ich)* ‘think/believe (I)’ furthermore have the special property to be prosodically integrated (Reis 2002).⁸ This sets them apart from other types of parentheticals that typically show comma intonation. Both properties in combination (flexibility in surface middle-field occurrence and prosodic integration) give rise to examples in which the critical expressions are ambiguous between an interpretation as V1 parentheticals or as SAdvS/MPs.

The most frequent verb in integrated V1 parentheticals is *glauben* ‘believe, think’ cf. (15a). Apart from its parenthetical uses, some speakers use the expression *glaub* (with drop of the 1st person subject pronoun) in spoken communication, cf. (15b).⁹ In such cases, especially those

⁸ Integrated V1-parentheticals are prosodically characterized by the absence of pauses, the absence of a separate focus-background-structure, and syntactically by V1 order (Reis 2002).

⁹ The examples in (15) are transcripts of a conversation between mother and daughter from the DGD database of recorded spoken language. Both speakers use prosodically integrated V1 parentheticals with the verb *glauben*, the mother with an overt 1st person subject pronoun (15a), the daughter without it (15b). – Parallel examples without comma or hyphen punctuation are sporadically attested also in written corpora, even with the drop of the pronoun.

with pronoun drop, *glaub* can be argued to be ambiguous between its original parenthetical use and a more recent use that shows some similarities with MPs (Viesel 2011).¹⁰

- (15) a. wraps sind **glaub** *ich* mein lieblingsessen aktuell
 wraps are believe I my favourite-food at.the.moment
 ‘Wraps are, I think, my favourite food at the moment’
 (DGD, FOLK_E_00331_SE_01_T_02, 1457)
- b. da sind **glaub** zwei drin in der packung
 there are believe two inside in the package
 ‘There are two in the package, I think/I suppose/presumably/allegedly.’
 (DGD, FOLK_E_00331_SE_01_T_02, 1184)

Also in the case of *scheints* discussed above, we find parallel examples with prosodic integration, in spoken corpora, and without comma or hyphen punctuation, in written texts (Axel-Tober 2016). There is a difference between *glaub* and *scheints*, though. As mentioned above, some German speakers can front *scheints* to the prefield. Sporadically, prefield placement of *scheints* is even attested in newspaper corpora (cf. section 1). *glaub*, however, is (as yet) not attested in the prefield, not even in corpora of the spoken language. Even speakers, who frequently use it with subject pronoun drop and prosodic integration, reject prefield positioning:

- (16) ***Glaub** ist das gar nicht so schlimm.
 think is that at.all not so bad
 ‘That is not so bad, I guess.’

Hence, Viesel (2011) comes to the conclusion that *glaub* is syntactically similar to a MP. Semantically, *glaub* is closer to SAdvS in that it can be argued to convey an epistemic meaning that influences the truth-values of a sentence (ibid. 159).

To sum up: In both paths the reanalysis is enabled by a syntactic surface ambiguity in combination with an interpretative ambiguity. In the path starting from low, i.e. event-related

¹⁰ German *bitte* ‘please’, sometimes analyzed as a MP, is another case originating from a parenthetical verb in the 1st person singular.

(17) [_{CP} ... *erbärmlicherweise/eigentlich* ... [_{AspP} ... *erbermlicher weise/eigentlich* ...]]



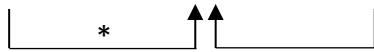
(18) [CP] > [CP ... *scheints/glaub* ...]
 [CP ... *scheints* ...]
 [CP ... *glaub* ...]

¹¹ The schematic representation in (17) abstracts away from differences in the syntactic behaviour of SAdv_s and MP_s. We do not subscribe to a particular cartographic analysis. In a Cinquean-style analysis, SAdv_s sit in dedicated functional projections in the left IP/TP-region (Mood_{speech act}P > Mood_{evaluative}P > Mood_{evidential}P > Mod_{epistemic}P). Event-related adverbs occur in the projection where verbal aspect is represented (AspP in (17)) or in deeper functional projections. In Krifka's (to appear) more recent cartography, (subjective) epistemic and evidential SAdv_s are related to the Judgement Phrase (JP) above TP and MP_s to the Act Phrase (ActP) at the leftmost clausal periphery. Event-related adverbs and adverbials are merged within the TP-domain and thus have an effect on the truth-conditional content of the proposition, which is syntactically encoded by the TP in Krifka's approach.

¹³ An anonymous reviewer raises the question whether biclausal, hypotactic structures of the type “It seems that.../ It may be that ...” could also serve as source constructions in the development of SAdvs. At least for German, these constructions are never ambiguous with adverbs and we correctly predict that German does not have any SAdvs with agglutinated complementizers. Even in the sample of European languages studied by Ramat & Ricca (1998) such adverbs are rarer (e.g. Lat. *forsitan* ‘perhaps’ < *fors sit* an ‘be the chance that’; Sp. (American, colloquial) *dizque* ‘allegedly’, *ibid.* p. 232) than those without complementizers. For Swedish *kanske* (*att*), Beijering & Norde (2019) point out that *kan ske* ‘can happen’ is also found parenthetically without the complementizer in Early Modern Swedish.

degrammaticalization in the sense that, syntactically, it would involve a lowering in the sentential hierarchy instead of an upward reanalysis, while, semantico-pragmatically, it would result in meaning enrichment rather than in meaning bleaching:

(19) *aber* [_{CP} ... *aber* ... [_{AspP} *aber* ...]]



We have demonstrated the two pathways only with examples from German. It should be noted, however, that both types of developments seem to represent two major developments in the sample of 41 European languages from different families studied in Ramat & Ricca (1998: 248).

4. Secondary grammaticalization?

Secondary grammaticalization is usually defined as the process that affects items that are already (to some degree) grammatical(ized) (Breban 2014). Assuming that SAdv results from a grammaticalization process, any further development of a SAdv might thus be called secondary grammaticalization. Below, two common notions within grammaticalization theory, namely weakening and syntactic climbing, will be discussed in this respect. Our findings suggest that there is a restriction that prevents SAdv from further grammaticalization yielding higher SAdv or MPs, as represented in (20).

- (20) a. *strong SAdv > weak SAdv (= MP)
 b. *lower SAdv > higher SAdv

This is surprising not only because these paths may be predicted from weakening and climbing processes, but also because these elements occupy surface positions that are very close to each other, which should result in a high degree of syntactic ambiguity.

4.1 From strong to weak SAdv?

In recent syntactic theory, MPs have been described synchronically as weak SAdvS (e.g. Grosz 2007).¹⁴ In this view, MPs are structurally impoverished in comparison to SAdvS.¹⁵ Diachronically, this suggests SAdvS as primary source category for the grammaticalization of MPs (cf. Coniglio 2022). The process of weakening involved would then cause the impoverishment of internal structure as well as the semantic ineffability typically associated with MPs. This, however, is not borne out by the facts. The first observation that can be made is that at least most MPs clearly do not originate from SAdvS. In the literature, SAdvS have only been proposed as one out of several source categories. The second observation is that for those MPs for which an origin from SAdvS has been suggested, this is questionable.

As shown in 2.1 (and in Coniglio 2022), these cases have a plausible origin in lower adverbs. Especially in the case of *vielleicht*, the alternative origin from the SAdv is in contrast highly implausible, since the SAdv and the MP exhibit a different distribution with regard to sentence type. SAdv *vielleicht* appears in declaratives as well as in yes/no questions, but not in exclamatives. MP *vielleicht*, on the other hand, appears only in exclamatives. Moreover, SAdv *vielleicht* denotes a possibility which has a weakening effect on the sentence, while exclamatives are known to be factive (Zanuttini & Portner 2003). This makes exclamatives incompatible with *vielleicht* as a SAdv. Hence, there is no pragmatically ambiguous context in which a reanalysis could have taken place.

So, while a grammaticalization path from a SAdv to a MP would be theoretically plausible and motivated, the observations above suggest that there is in fact a restriction against this path. Secondary grammaticalization, in the sense that an already grammaticalized SAdv is grammaticalized further into a MP through a process of weakening, does not seem to take place (20a). Since the origin of many SAdvS and MPs can be traced back to similar sources, i.e. lower adverbs and parentheticals, our findings further suggest that MPs are grammaticalization alternatives to SAdvS into the discourse (*ja*, *doch*) or modal (*wohl*, *glaub*) domains.

4.2 Syntactic climbing within higher adverbial domains?

¹⁴ Crosslinguistically, this leads to the question whether SAdvS are really more common than MPs, as a reviewer suggests, or whether rather this is not a meaningful distinction in the first place.

¹⁵ This fits also their respective degree of non-at-issueness, a characteristic which is uncontroversial for MPs, but sometimes disputed for some SAdv: in discourse, SAdv cannot be rejected easily in comparison to other adverbials, but more easily in comparison to MPs (Müller 2021). Thus, a path from SAdv to MP would also fit the notion of grammaticalization proposed by Boye & Harder (2012), which is based on the conventionalization of discursively secondary status.

Another aspect which is often associated with grammaticalization is reanalysis up the tree (Roberts & Roussou 2003). Cartographic research has proposed fine-grained functional hierarchies that relate different subcategories of SAdvS to different structural positions in the clausal spine (cf. Cinque 1999 and fn. 11). Examples for typical categories are epistemic adverbs (*vielleicht* ‘maybe’), evidential adverbs (*offenbar* ‘apparently’), evaluative adverbs (*erbärmlicherweise* ‘pitifully’, *leider* ‘unfortunately’) and speech act adverb(ial)s (*ehrlich* (*gesagt*) ‘frankly’, *kurzum* ‘in short’). The functional projections hosting these adverbs are ordered from low to high as represented in (21):

(21) epistemic > evidential > evaluative > speech act

In a fine-grained syntactic analysis, the upward reanalysis described in sections 2.1 and 3 may target one of categories in (21). At least regarding German, however, there is to our knowledge not a single development along this hierarchy. This suggests that, after an adverb has entered the higher functional domain associated with speaker orientation, it is cut off from further development. Secondary grammaticalization, in the sense that an already grammaticalized SAdv is grammaticalized further into a different SAdv higher up the functional hierarchy, does not seem to take place (20b).

5. Conclusion

We proposed a unified syntactic analysis that explains the grammaticalization of SAdvS and MPs as a development from the lower lexical layer or from parentheticals into weak and strong elements of the higher clausal functional domains. We focussed on the restriction of syntactic ambiguity that allows for both proposed paths, but rules out other paths proposed in the literature, namely the development from conjunctions and subordinated structures. We observed that, once an element has reached the higher functional domain, a further restriction rules out secondary grammaticalization. So far, this restriction remains unexplained and must be left for further research.

Corpora

DeReKo = Deutsches Referenzkorpus / Archiv der Korpora geschriebener Gegenwartssprache.
 Mannheim: Institut für Deutsche Sprache. <https://www.ids-mannheim.de/digspra/kl/projekte/korpora/>

DGD = Datenbank für Gesprochenes Deutsch. Mannheim: Institut für Deutsche Sprache.
https://dgd.ids-mannheim.de/dgd/pragdb.dgd_extern.welcome

DTA = Digitaler Teil des Deutschen Textarchivs. Berlin-Brandenburgische Akademie der Wissenschaften Deutsches Textarchiv. <http://deutschestextarchiv.de>

MWB = Mittelhochdeutsches Wörterbuch. <http://www.mhdwb-online.de/index.html>

References

- Abendroth Scherf, Nathalie Katharina. 2018. Swedish modal particles. Analyses of *ju*, *väl*, *nog* and *visst*. PhD thesis, Humboldt-Universität zu Berlin.
- Alexiadou, Artemis. 1997. *Adverb placement: A case study in antisymmetric syntax*. Amsterdam: John Benjamins.
- Autenrieth, Tanja. 2002. *Heterosemie und Grammatikalisierung bei Modalpartikeln*. Tübingen: Niemeyer.
- Axel-Tober, Katrin. 2016. Satzadverbale im Deutschen: synchrone und diachrone Fragen bei einem ‚scheints‘ alten Thema. In Sergio Neri, Roland Schuhmann & Susanne Zeilfelder (eds.), *"dat ih dir it nu bi huldi gibu" – Linguistische, germanistische und indogermanistische Studien Rosemarie Lühr gewidmet*, 23–33. Wiesbaden: Reichert.
- Axel-Tober, Katrin & Kalle Müller. 2017. Evidential adverbs in German. Diachronic development and present-day meaning. *Journal of Historical Linguistics* 7 (1/2). 9–47.
- Beijering, Karin & Muriel Norde. 2019. Adverbial semi-insubordination constructions in Swedish: Synchrony and diachrony. In Karin Beijering, Gunther Kaltenböck & María S. Sansiñena (eds.), *Insubordination: Theoretical and empirical issues*, 79–106. Berlin: De Gruyter.
- Berry, James. 2011. Diachronic adverbial morphosyntax: A minimalist study of lexicalization and grammaticalization. PhD thesis, Arizona State University.
- Boye, Kasper & Peter Harder. 2012. A usage-based theory of grammatical status and grammaticalization. *Language* 88 (1). 1–44.
- Breban, Tine. 2014. What is secondary grammaticalization? Trying to see the wood for the trees in a confusion of interpretations. *Folia Linguistica* 48 (2). 469–502.

- Burkhardt, Armin. 1994. Abtönungspartikeln im Deutschen: Bedeutung und Genese. *Zeitschrift für germanistische Linguistik* 22. 129–151.
- Cinque, Guglielmo. 1999. *Adverbs and functional heads. A cross-linguistic perspective*. Oxford: OUP.
- Coniglio, Marco. 2011. *Die Syntax der deutschen Modalpartikeln. Ihre Distribution und Lizenzierung in Haupt- und Nebensätzen*. Berlin: Akademie Verlag.
- Coniglio, Marco. 2022. On the adverbial origin of German modal particles. In Xabier Artiagoitia, Arantzazu Elordieta & Sergio Monforte (eds.), *Discourse particles: Syntactic, semantic, pragmatic and historical aspects*, 13–40. Amsterdam: John Benjamins.
- Detges, Ulrich & Richard WALTEREIT. 2016. Grammaticalization and pragmaticalization. In Susann Fischer & Christoph Gabriel (eds.), *Manual of grammatical interfaces in Romance*, 635–658. Berlin: De Gruyter.
- Diewald, Gabriele. 2011. Pragmaticalization (defined) as grammaticalization of discourse functions. *Linguistics* 49 (2). 365–390.
- Elsner, Daniela. 2015. Adverbial morphology in German: Formations with–weise/–erweise. In Karin Pittner, Daniela Elsner & Fabian Barteld (eds.), *Adverbs. Functional and diachronic aspects*, 101–132. Amsterdam: John Benjamins.
- Fischer, Olga. 2011. Grammaticalization as analogically driven change? In Bernd Heine & Heiko Narrog (eds.), *The Oxford handbook of grammaticalization*, 3–42. Oxford: OUP.
- Frey, Werner & Karin Pittner. 1998. Zur Positionierung der Adverbiale im deutschen Mittelfeld. *Linguistische Berichte* 176. 489–534.
- Giorgi, Alessandra & Fabio Pianesi. 2005. Credo (I believe): Epistemicity and the syntactic representation of the speaker. *University of Venice Working Papers in Linguistics* 15. 105–152.
- Grosz, Patrick. 2007. German discourse particles are weak sentence adverbs. In Cecilia Picchi & Alan Pona (eds.), *Proceedings of the XXXII Incontro di Grammatica Generativa*, 79–91. Edizioni dell'Orso.
- Gutzmann, Daniel. 2015. *Use-conditional meaning. Studies in multidimensional semantics*. Oxford: OUP.
- Harris, Alice C. & Lyle Campbell. 1995. *Historical syntax in cross-linguistic perspective*. Cambridge: CUP.
- Haspelmath, Martin. 1998. Does grammaticalization need reanalysis? *Studies in Language* 22. 315–351.
- Hetland, Jorunn. 1992. *Satzadverbien im Fokus*. Tübingen: Narr.

- Hopper, Paul & Elizabeth Closs Traugott. 1993. *Grammaticalization*. Cambridge: CUP.
- Jackendoff, Ray. 2002. *Foundations of language. Brain, meaning, grammar, evolution*. Oxford: OUP.
- Krifka, Manfred (to appear): Layers of assertive clauses: Propositions, judgements, commitments, acts. In Jutta Hartmann & Angelika Wöllstein (eds.), *Propositionale Argumente im Sprachvergleich: Theorie und Empirie*. Tübingen: Narr.
- Lehmann, Christian. 1995. *Thoughts on grammaticalization*. München: LINCOM.
- Müller, Kalle. 2021. *Satzadverbien und Evidentialität*. Berlin: De Gruyter.
- Paraschkewoff, Boris. 1976. Zur Entstehung und Entwicklungsgeschichte der Bildungen auf ‚-weise‘. *Beiträge zur Geschichte der deutschen Sprache und Literatur* 97. 165–211.
- Pittner, Karin. 2015. Between inflection and derivation. Adverbial suffixes in English and German. In Karin Pittner, Daniela Elsner & Fabian Barteld (eds.), *Adverbs. Functional and diachronic aspects*, 133–156. Amsterdam: John Benjamins.
- Potts, Christopher. 2005. *The logic of conventional implicatures*. Oxford: OUP.
- Ramat, Paolo & Davide Ricca. 1998. Sentence adverbs in the languages of Europe. In Johan van der Auwera (ed.), *Adverbial constructions in the languages of Europe* (EALT 20:3). Berlin: Mouton de Gruyter, 187–275.
- Reis, Marga. 2002. Wh-movement and integrated parenthetical constructions. In Jan-Wouter Zwart & Werner Abraham (eds.), *Studies in comparative Germanic syntax. Proceedings from the 15th workshop on comparative Germanic syntax* (Linguistics Today 53), 3–40. Amsterdam: John Benjamins.
- Remberger, Eva-Maria. 2020. Discourse and pragmatic markers in the Romance languages. In Michele Loporcaro & Francesco Gardani (eds.), *The Oxford encyclopedia of Romance linguistics*. Oxford: OUP.
- Roberts, Ian & Anna Roussou. 2003. *Syntactic change. A minimalist approach to grammaticalization*. Cambridge: CUP.
- Swan, Toril. 1988. Sentence adverbials in English: A synchronic and diachronic investigation. Oslo: Novus.
- Thompson, Sandra A. & Anthony Mulac. 1991. *A quantitative perspective on the grammaticalization of epistemic parentheticals in English*. In Elizabeth Closs Traugott & Bernd Heine (eds.), *Approaches to grammaticalization. Volume II. Types of grammatical marker*, 313–330. Amsterdam: John Benjamins.
- Thurmair, Maria. 1989. *Modalpartikeln und ihre Kombinationen*. Tübingen: Niemeyer.

- Traugott, Elizabeth C. 1989. On the rise of epistemic meanings in English: An example of subjectification in semantic change. *Language* 65. 31–55.
- Traugott, Elizabeth C. 2010. Revisiting subjectification and intersubjectification. In Kristin Davidse, Lieven Vandelanotte & Hubert Cuyckens (eds.), *Subjectification, intersubjectification and grammaticalization*, 29–70. Berlin: Mouton De Gruyter.
- Traugott, Elizabeth C. 2011. Grammaticalization and mechanisms of change. In Bernd Heine & Heiko Narrog (eds.), *The Oxford handbook of grammaticalization*, 19–30. Oxford: OUP.
- van Gelderen, Elly. 2011. *The linguistic cycle: Language change and the language faculty*. Oxford: OUP.
- Viesel, Yvonne. 2011. 'glaubt er', 'glaub ich', 'glaub': Integrierte V1-Parenthesen, Extraktion aus V2-Komplementen, Grammatikalisierung. *Linguistische Berichte* 226. 129–169.
- Wegener, Heide. 2002. The evolution of the German modal particle *denn*. In Ilse Wischer & Gabriele Diewald (eds.), *New reflections on grammaticalization*, 379–394. Amsterdam: John Benjamins.
- Zanuttini, Raffaella & Paul Portner. 2003. Exclamative clauses: At the syntax-semantics interface. *Language* 79. 39–81.

A

adverbial phrase 3, 5, 6

E

eigentlich 1, 2, 4, 5, 12

epistemic 5, 6, 7, 10, 11, 12, 15

-erweise 3, 4, 9

event-related adverb 3, 4, 8, 11, 12

evidential 6, 7, 10, 12, 15

G

grammaticalization 1, 2, 3, 5, 6, 8, 14, 15

H

higher functional domains 9, 12, 15

L

lower adverb 3, 4, 10, 12, 14

M

manner adverbial 3, 4, 8, 9, 10

modal particles 1–15

P

parenthetical sentences 3, 6, 7, 8, 10, 11, 12, 15

path 3, 4, 5, 6, 8, 11, 12, 13, 14, 15

R

reanalysis 3, 6, 7, 8, 10, 11, 12, 13, 14, 15

S

scheints 6, 7, 11, 12

secondary grammaticalization 13, 14, 15

sentence adverbs 1–15

syntactic ambiguity 7, 8, 10, 12, 13, 14

syntactic climbing 13, 14

T

temporal adverbs 3, 4, 7

V

vielleicht 1, 2, 5, 6, 7, 14, 15